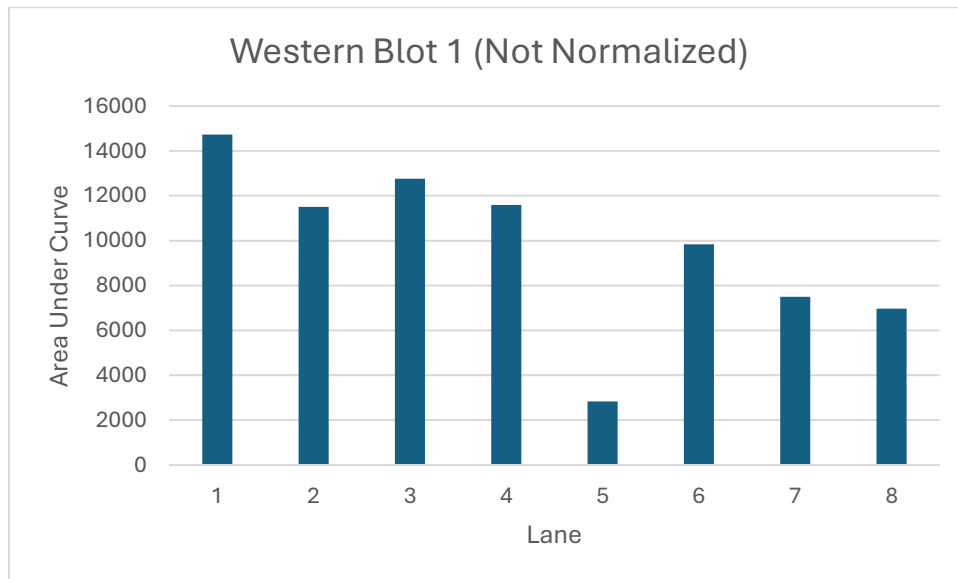


CM515: Intro to Image Processing & Analysis Assignment

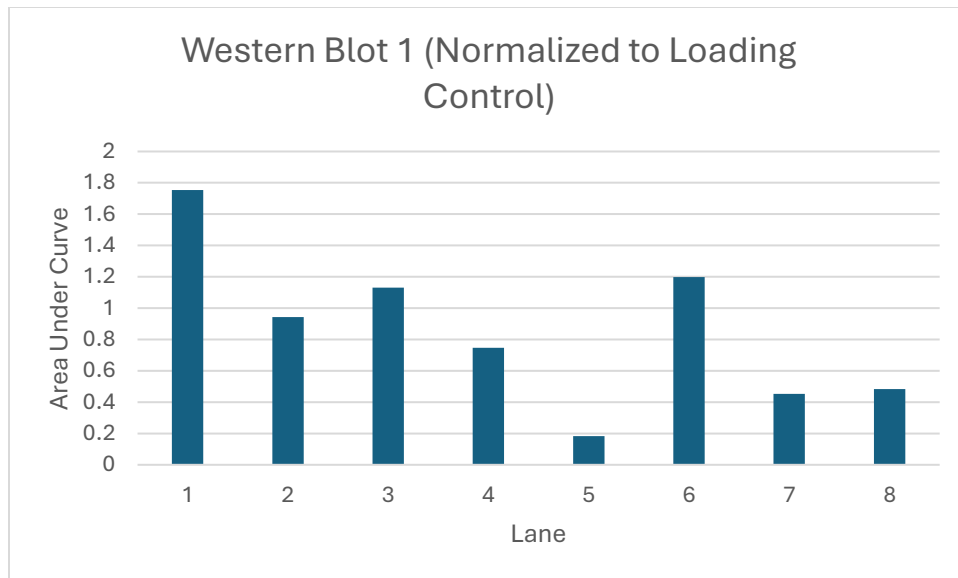
Part 1:

Exercise 1: Quantify a Western Blot

- Open "WesternBlot1.tif" from the Image Bank folder and open it in FIJI
- Using the guide in the lecture slides, quantify the bands.
- In your favorite plotting software, create a bar plot showing the volume of each band, and paste it here.



- Next, open "Western1_LoadingControl.tif"
- Quantify the bands
- Divide your quantification of each lane from the first Western Blot by the quantification of the same lane in the Loading control blot. Use this value to create a plot showing the values of the bands normalized to the loading control, and paste it here



Exercise 2: Record a Macro

- Navigate to *Plugins > Macros > Record* (or use keyboard shortcut *Cmd/Ctrl + U*)
 - o Dialog box will pop up, and record all the actions you do
- Open an image from the provided sample images in FIJI (*File > Open Samples*), from one of the databases listed on the last slide, or from your own work/ a colleague's
- Make 2 modifications to the image (whatever you want! Adjust brightness contrast, change image type, add scale bar, etc!), and then close it.
- Copy & paste the record and the final product image here:

```
run("Tree Rings");
```

```
//run("Brightness/Contrast...");
```

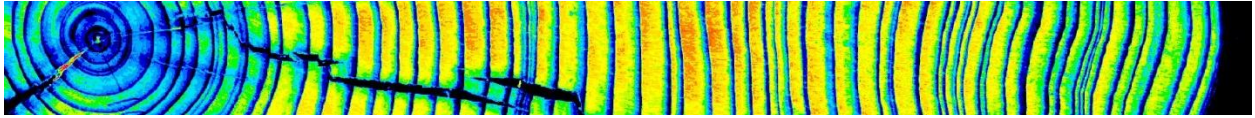
```
run("Apply LUT");
```

```
run("Close");
```

```
run("16 colors");
```

```
saveAs("Jpeg", "C:/Users/kmark/Documents/GitHub/CM515-2026/modules/wk_10_image_analysis/Tree_Rings_Modified.jpg");
```

```
close;
```



- Bonus (not required but fun!): Make this record into a macro that can be applied to another image!